

Quiz 2 Solutions: Applied Natural Language Processing DATA 441/641

1. What is the main limitation of the Bag of Words (BoW) model?
 - (a) **It does not capture word order.**
 - (b) It assigns the same vector to all documents.
 - (c) It ignores semantic meaning and context.
 - (d) It cannot be applied to NLP tasks.
2. How does CBOW in Word2Vec generate word embeddings?
 - (a) It predicts the next word given previous words.
 - (b) **It predicts a word given its surrounding words.**
 - (c) It assigns weights to words based on TF-IDF.
 - (d) It clusters words based on their occurrence in sentences.
3. Which of the following statements is true about Skip-gram compared to CBOW?
 - (a) **Skip-gram predicts surrounding words given a target word.**
 - (b) Skip-gram is faster to train than CBOW.
 - (c) Skip-gram is used only for document classification.
 - (d) Skip-gram and CBOW both require hierarchical softmax to function.
4. What is the key idea behind the GloVe embedding model?
 - (a) It learns embeddings by predicting words from context.
 - (b) **It uses co-occurrence statistics to learn word representations.**
 - (c) It applies deep learning to generate one-hot encodings.
 - (d) It uses frequency counts directly for similarity.
5. Which embedding method is least effective in handling out-of-vocabulary (OOV) words?
 - (a) **One-hot encoding**
 - (b) Subword-based embeddings (FastText)
 - (c) Word2Vec
 - (d) GloVe
6. Which of the following best describes a challenge when training word embeddings?
 - (a) **Large vocabulary size can make training computationally expensive.**
 - (b) CBOW and Skip-gram require word frequency equalization.
 - (c) Word embeddings do not improve semantic understanding.

- (d) Word embeddings must be retrained every time a new document is added.
7. Which of the following best describes negative sampling in Word2Vec?
- (a) **A method to approximate softmax by updating a subset of word probabilities.**
 - (b) A way to remove low-frequency words from training.
 - (c) A technique used to balance class distribution in word embeddings.
 - (d) A modification that forces Skip-gram to predict words in a predefined order.
8. What is the architecture of the CBOW model?
- (a) A CNN-based deep learning model for text processing.
 - (b) A recurrent neural network that generates text sequences.
 - (c) A transformer-based encoder that learns sentence embeddings.
 - (d) **A two-layer neural network that predicts a word given surrounding words.**
9. Which of the following best describes the architecture of Skip-gram?
- (a) A network that uses hierarchical softmax to classify words into clusters.
 - (b) A deep LSTM-based model for contextual word representation.
 - (c) A network that constructs word embeddings using character n-grams.
 - (d) **A network that takes one word as input and predicts multiple surrounding words.**
10. Suppose you have a vocabulary of V unique words and you decide to represent text sequences using n -grams of length n . What is the maximum possible vocabulary size in this case?
- (a) $V + n$
 - (b) V^n **is correct!**
 - (c) nV
 - (d) n^V
11. **(641 only)** In Word2Vec training, what impact does increasing the window size have on the resulting word embeddings?
- (a) It makes embeddings more syntactic rather than semantic.
 - (b) **It captures broader contextual relationships, improving semantic information.**
 - (c) It forces the embeddings to be more dependent on frequency counts.
 - (d) It has no impact on the learned word representations.
12. **(641 only)** In Skip-gram with negative sampling, how is the loss function modified compared to traditional softmax-based training?
- (a) It introduces an additional regularization term to control overfitting.
 - (b) It computes cosine similarity between the target and context word vectors instead of probability distributions.
 - (c) **It replaces the probability calculation over the entire vocabulary with binary classification for sampled words.**
 - (d) It ensures that every word pair in the dataset contributes equally to gradient updates.